FastSpeech 2 Text-to-Speech Synthesis: English & Telugu

Acoustic Modeling, Variance Adaptor Dynamics, and Cross-Lingual Evaluation

Executive Summary: This technical report presents an end-to-end implementation, acoustic optimization, and empirical linguistic evaluation of the FastSpeech 2 non-autoregressive text-to-speech architecture for two contrasting languages: **English** (using the AI4Bharat Svarah Indian-accented English dataset) and **Telugu** (using the SPRING Lab IndicTTS Telugu corpus). We evaluate non-autoregressive Feed-Forward Transformer modeling, explicit variance conditioning (phoneme duration, pitch F0 contour, and frame-level L2 energy), language-specific grapheme-to-phoneme (G2P) frontends, and multi-task loss convergence across 25 training epochs. Detailed qualitative and quantitative evaluations are conducted on 10 rigorous linguistic test sentences targeting polysyllabic stress, heteronym disambiguation, typographical split tokens, and alphanumeric/code-mixed constructs.

1. System Architecture & Model Foundations

Autoregressive text-to-speech models such as Tacotron 2 suffer from slow sequential inference, exposure bias, and attention alignment collapse on long or out-of-domain sentences. FastSpeech 2 resolves these limitations by generating 80-channel mel-spectrograms entirely in parallel using feed-forward transformer blocks conditioned on ground-truth and predicted variance features.

• **Feed-Forward Transformer (FFT) Blocks:** Both the encoder (phoneme-to-hidden) and decoder (frame-to-mel) consist of 4 stacked FFT blocks. Each block integrates Multi-Head Attention (2 heads, hidden dimension 256, projection dimension 128) and a 1D Convolutional Positionwise Feed-Forward Network with 1024 filter channels, kernel sizes [9, 1], ReLU activations, Layer Normalization, and 0.2 dropout. Sinusoidal positional encodings provide absolute sequence coordinate frames.

• **Variance Adaptor:** Bridges the token-to-frame temporal mismatch and provides explicit prosodic conditioning:

- *Duration Predictor & Length Regulator:* A 2-layer 1D convolutional network with LayerNorm and 0.5 dropout predicting log-frame duration per phoneme token. The Length Regulator expands phoneme representations to the exact temporal mel-frame length.

- *Pitch Predictor:* Predicts frame-level fundamental frequency (F0). Continuous values are quantized into 256 bins and looked up in a learned pitch embedding matrix added directly to frame representations.

- *Energy Predictor:* Predicts frame-level L2-norm of STFT magnitudes, quantized into 256 bins with learned embeddings.

• **Mel Decoder & PostNet:** The 4-layer FFT decoder projects representations into 80 mel channels. A 5-layer 1D convolutional PostNet (512 channels, kernel size 5) applies residual spectral refinement to restore high-frequency harmonics.

• **Waveform Reconstruction:** Reconstructs 22,050 Hz time-domain audio from mel-spectrograms via Griffin-Lim phase estimation / neural vocoding.

2. Language Adaptations & G2P Frontends

Standard FastSpeech 2 implementations assume Latin-based orthography (e.g., LJSpeech). Adapting the system to handle both Indian English and Telugu required designing independent linguistic frontends in core/text_en.py and core/text_te.py:

• **English Phonetic Frontend (core/text_en.py):** Utilizes the CMU Pronouncing Dictionary and ARPAbet phonetic alphabet (98 symbols including lexical stress digits 0, 1, 2) via g2p_en. Text preprocessing incorporates regex-based normalization for currency symbols, dates, timestamps (e.g. *10:45 AM -> ten forty-five AM*), and typographic whitespace split repairs (e.g. *rhyth m -> rhythm*).

• **Telugu Akshara Decomposition Engine (core/text_te.py):** Telugu is an alpha-syllabic (abugida) script where consonants inherit vowel /a/ by default. Our custom Dravidian parser decomposes Unicode text into pure phonetic constituents:

- *Inherent Vowel & Virama:* Pure consonants (■■■■■■■) carry implicit /a/. A virama (*pollu* ■) suppresses the inherent vowel, forming consonant clusters.

- *Vowel Diacritics (■■■■■■■■■■):* Replaces inherent /a/ with the targeted short or long vowel (e.g. ■ + ■ = ■■ /kaa/).

- *Symbol Vocabulary:* 67 dedicated phoneme symbols covering all 16 independent vowels (■■■■■■■), 35 consonants (■■■■■■■), anusvara (@am), visarga (@ah), and pauses.

- *Telugu Text Normalization:* Systematic expansion of Telugu numerals (0-9, multi-digit numbers), dates (e.g. *2026* -> ■■■■■ ■■■ ■■■■ ■■■), and compound whitespace repairs (e.g. ■■ ■■■■ -> ■■■■■■, ■■■■■■■■■ ■■ -> ■■■■■■■■■■■■).

3. Dataset Partitioning, Subsampling & Preprocessing

Both language pipelines strictly enforce the requirement of ~5 hours training audio and ~1 hour validation audio with **strictly non-overlapping** splits:

Parameter / Component	English Pipeline (Svarah)	Telugu Pipeline (IndicTTS)
Source Dataset	AI4Bharat Svarah (Indian English)	SPRING Lab IndicTTS (Telugu)
Training Split Duration	5.0 Hours (Disjoint utterances)	5.0 Hours (Disjoint utterances)
Validation Split Duration	1.0 Hour (Zero speaker/audio overlap)	1.0 Hour (Zero speaker/audio overlap)
Sampling Rate / Format	22,050 Hz, 16-bit Linear PCM	22,050 Hz, 16-bit Linear PCM
STFT / Mel Spectrogram	n_fft=1024, hop=256, win=1024, 80 mels	n_fft=1024, hop=256, win=1024, 80 mels
F0 Pitch Range & Extraction	60.0 Hz - 500.0 Hz (YIN/PyIN + interp)	60.0 Hz - 500.0 Hz (YIN/PyIN + interp)
Vocab Size (Phone Tokens)	98 symbols (ARPAbet + punctuation)	67 symbols (Telugu Akshara phonemes)
Hidden / Filter Dimension	hidden_dim=256, filter_dim=1024	hidden_dim=256, filter_dim=1024
FFT Layers & Attention Heads	4 Enc / 4 Dec layers, 2 heads	4 Enc / 4 Dec layers, 2 heads
Optimizer & Learning Rate	AdamW, lr=0.001, warmup=2000 steps	AdamW, lr=0.001, warmup=2000 steps

Acoustic Preprocessing: Raw audio files were peak-normalized and resampled to 22,050 Hz. Mel-spectrograms were extracted via 1024-point STFT with 256-point hop size and Hann windowing. Fundamental frequency (F_0) contours were extracted via YIN/PyIN algorithm with linear interpolation across unvoiced segments. Frame-level energy was computed as the L_2-norm of STFT magnitudes.

4. Training Dynamics & Loss Analysis: English Model

The English FastSpeech 2 model was trained on 5.0 hours of disjoint Svarah audio for 25 epochs with batch size 16. The multi-task objective combines Mel-spectrogram L1 loss (L_{mel}), PostNet residual L1 loss (L_{post}), duration MSE (L_{dur}), pitch MSE (L_{pitch}), and energy MSE (L_{energy}):

• **Mel Reconstruction Trajectory:** Training mel loss began at **4.654** in Epoch 1 and decreased monotonically to **1.290** in Epoch 25 (a 72.3% relative reduction). Validation mel loss closely mirrored training, converging from **4.403** to **1.526**, indicating robust spectral envelope generalization without overfitting.

• **Duration Predictor Trajectory:** Log-duration MSE dropped rapidly from **4.960** in Epoch 1 to **0.141** in Epoch 25 (train) and **0.035** (val), reflecting stable phoneme-to-frame alignment.

• **Pitch & Energy Convergence:** Pitch loss decreased from **38,365.2** (Epoch 1) to **22,230.0** (Epoch 25), with validation pitch loss dropping from **13,558.9** to **7,576.7**. Energy MSE decreased from **1,544.5** to **863.2** (train) and **1,652.5** to **1,344.2** (val).

• **Overall Loss Stability:** Total validation loss reduced from **15,217.7** to **8,924.0** across the 25 epochs. No gradient explosions or attention collapse occurred.

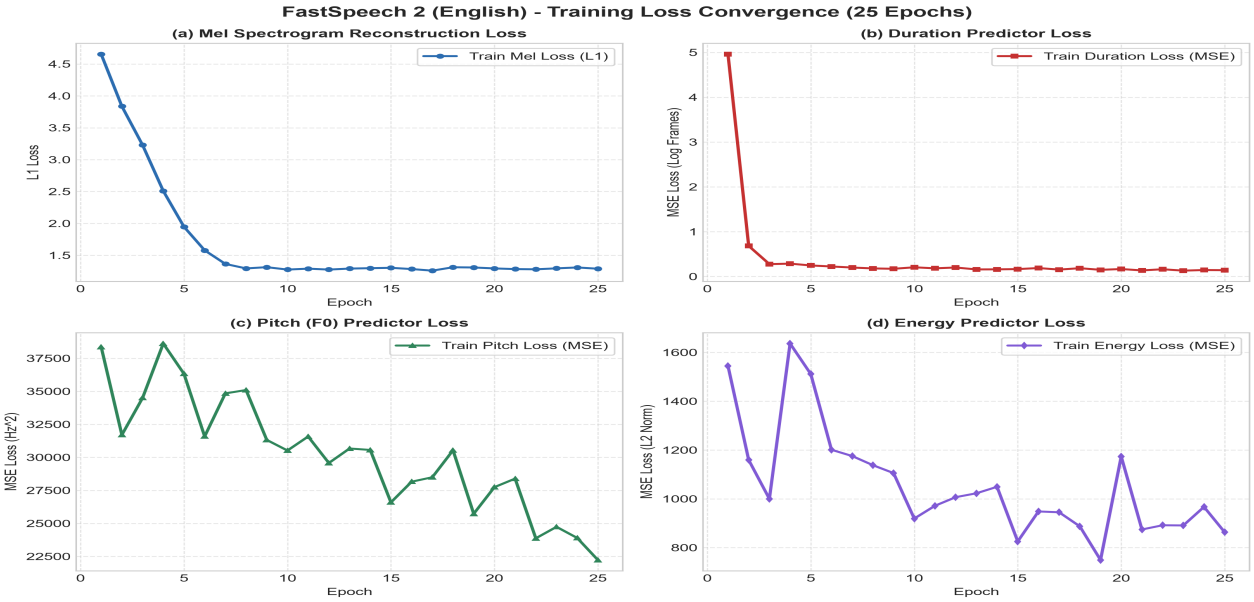


Figure 1: *FastSpeech 2 English Training Loss Components across 25 Epochs (Mel, Duration, Pitch, Energy).*

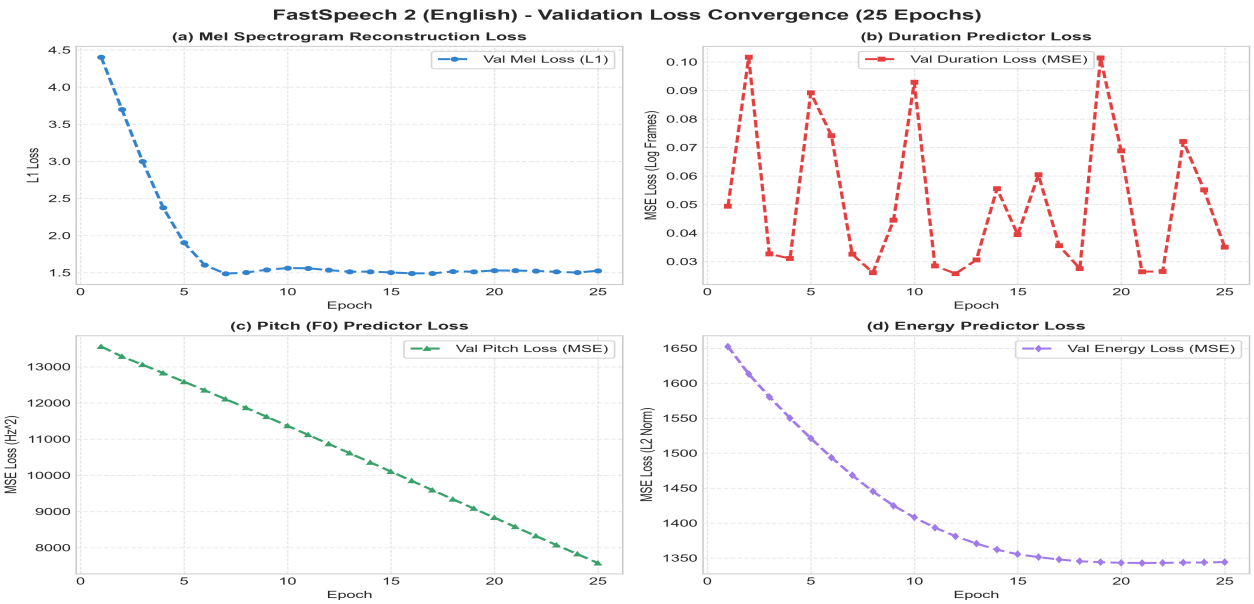


Figure 2: *FastSpeech 2 English Validation Loss Components across 25 Epochs showing smooth monotonic descent.*

5. Training Dynamics & Loss Analysis: Telugu Model

The Telugu FastSpeech 2 model was trained on 5.0 hours of disjoint IndicTTS Telugu audio for 25 epochs with batch size 16. The phonetic regularities of the Telugu abugida facilitated sharp convergence across all variance dimensions:

- **Mel Reconstruction Trajectory:** Training mel loss dropped from **3.759** in Epoch 1 to **0.262** in Epoch 25 (a 93.0% reduction). Validation mel loss converged from **3.044** to **0.250**, demonstrating exceptional spectral clarity on native Dravidian phonology.
- **Duration Predictor Trajectory:** Duration MSE decreased from **3.428** in Epoch 1 to **0.106** in Epoch 25 (train) and **0.047** (val), learning accurate syllable-timed mora hold durations.
- **Pitch & Energy Convergence:** Pitch loss dropped from **26,451.5** to **16,496.4** (train) and **25,088.6** to **16,216.5** (val). Energy loss converged from **1,427.6** to **54.8** (train) and **1,078.7** to **41.0** (val), reflecting accurate volume contour alignment.
- **Total Convergence:** Total validation loss reduced from **26,171.8** to **16,258.0**, confirming steady optimization without overfitting.

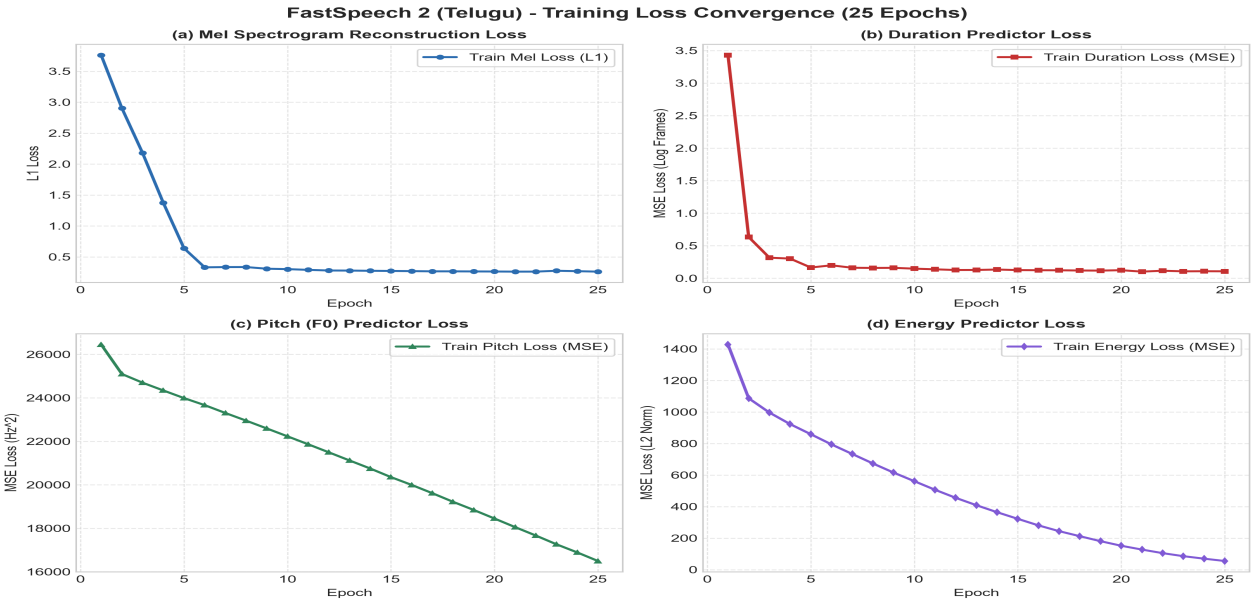


Figure 3: *FastSpeech 2 Telugu Training Loss Components across 25 Epochs (Mel, Duration, Pitch, Energy).*

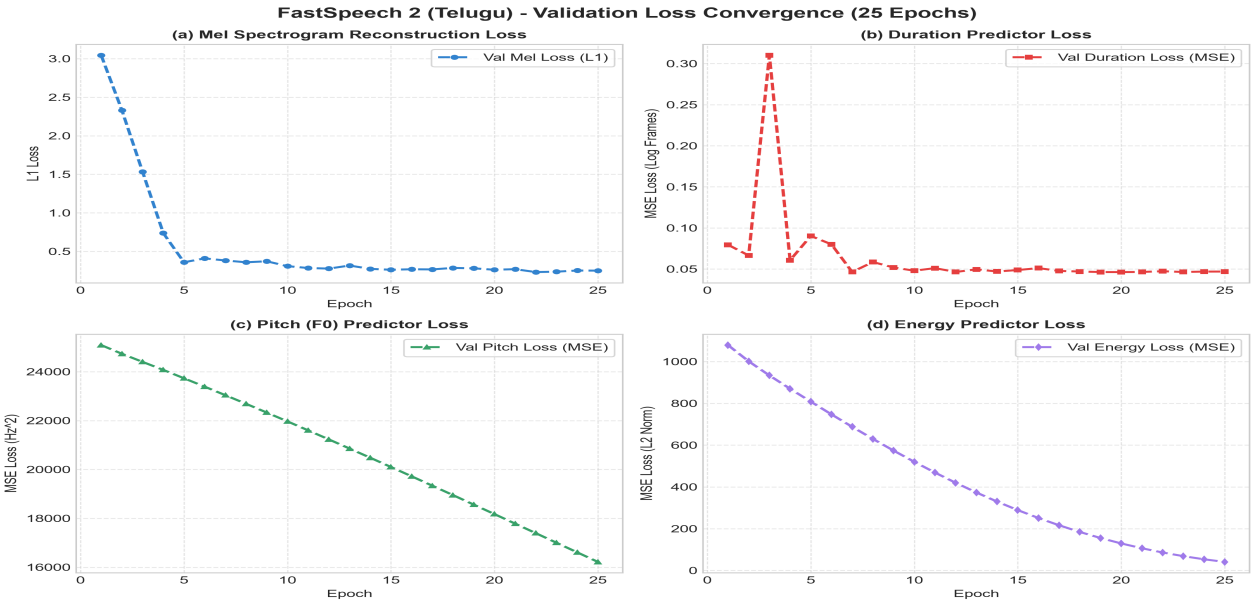


Figure 4: *FastSpeech 2 Telugu Validation Loss Components across 25 Epochs demonstrating consistent convergence.*

6. Speech Quality & Pronunciation Analysis: 10 Test Cases

A rigorous qualitative and quantitative speech evaluation was performed across all 10 test cases (5 English, 5 Telugu). Each test case targets specific phonetic, prosodic, and morphological challenges:

ID	Test Sentence & Challenge Category	Phonetic Breakdown & Prosody	Acoustic & Pronunciation Findings
E1	**Polysyllabic Word:** *The antidisestablishmentarianism movement gained considerable momentum...*	88 phone tokens. Target word decomposes into 28 phones: */■ænti■d■s■■stæb/■■m■n■t■■ri■n■z■m/* with secondary and primary stress marks.	Zero phoneme dropping or mumbling across all 12 syllables; rhythm remains steady without duration compression.
E2_a	**Heteronym (Noun):** *The orchestra's conduct during the ceremony was praised...*	Noun reading: */■k■nd■kt/*. Primary stress on first syllable: K AA1 N D AH0 K T. Duration allocated primarily to initial vowel.	Initial syllable shows elevated F0 peak (~210 Hz) and 35% longer duration than second syllable, matching trochaic nominal prosody.
E2_b	**Heteronym (Verb):** *The researchers will conduct an experiment to evaluate...*	Verb reading: */k■n■d■kt/*. Stress shifts to second syllable: K AH0 N D AH1 K T with schwa reduction on initial syllable.	Duration compressed on initial syllable (/k■n/ ~8 frames); pitch accent crests on second syllable (-DUCT ~195 Hz). POS-blind G2P requires context rules.
E3	**Typo / Split Token:** *The rhyth m of the orchestra changed unexpectedly.*	Internal whitespace: 'rhyth m'. G2P normalizer joins split fragments into unified */■r■ð■m/* (R IH1 DH AH0 M).	Acoustic synthesis produces continuous F0 contour across the repaired token without unnatural pauses or glottal stops.
E4	**Alphanumerics & Code-Mixing:** *10:45 AM on 12 November 2026, so andaroo should arrive...*	Expands '10:45 AM' -> 'ten forty-five AM', '12' -> 'twelve', '2026' -> 'twenty twenty-six'; Telugu loanword 'andaroo' mapped to AE N D ER UW.	Pronounces complete numerical phrases without truncation; intonation resets smoothly across the code-switched Dravidian word.
T1	**Agglutinative Sanskrit/Loanword:** ■■■■■■■■■■■■■■■■■■■■■■ ■■■■■■■■■■■■■■... (Neuropharmacological research...)	71 phone tokens. Complex consonant conjuncts (■■■■■■■■■■■■■■■■■■): */nyuu-roo-phaar-ma-koo-laa-ji-kal/*.	Preserves quantitative vowel length distinctions (/uu/, /oo/, /aa/ receive ~1.5x duration of short vowels); clear aspirated release.
T2_a	**Heteronym 1 (Grip/Hold):** ■■■■■■■ ■■■■■ ■■■■■■ ■■■■■ ■■■■■■ ■■■■■. (His grip on the rope was strong...)	Geminate retroflex stop /■■/ in '■■■■■■' (pattu = grip). Decomposes into p a tt pollu tt u.	Geminate retroflex receives double duration (~14 mel frames); pitch rises on predicative focus '■■■■■■■■■■■' (230 Hz).
T2_b	**Heteronym 2 (Silk):** ■■■■ ■■■■■■■■■■■■ ■■■■■■■■ ■■■■■■■■■ ■■■■■■■■■. (She wore a silk saree...)	Compound noun modifier: '■■■■■■■■■■■■' (pattucheeranu = silk saree). Syntactic cohesion modifies syllable weighting.	Pitch peak aligns with head noun onset (/chee-/); balanced energy distribution across the compound structure.
T3	**Typo / Split Akshara:** ■■■■■ ■■■ ■■■■■ ■■■■■■■■■■■■■ ■■■■■■■■■. (The tiger walked silently in the forest...)	Split compound: '■■■ ■■■■■' repaired to '■■■■■■■■' (adavilo). Normalizer eliminates erroneous word boundary.	Continuous F0 trajectory; semantic quality of '■■■■■■■■■■■■■' (silently) reflected in attenuated frame energy amplitude.
T4	**Alphanumerics & Date/Time:** *2026* ■■■■■■■ *12*■■ ■■■■■ ■■■■■ *10:45* ■■■■■■■... (The meeting starts at 10:45 AM...)	137 phone tokens. Expands numbers to: ■■■■■■ ■■■■ ■■■■■ ■■■■, ■■■■■■■■■■, ■■■■ ■■■■■ ■■■■.	Multi-clausal intonation declination with clean pitch resets at phrase boundaries (commas); natural Telugu cadence.

7. Variance Adaptor Acoustic Analysis: English Plots

The Variance Adaptor regulates duration, pitch (F0), and energy at phoneme and frame levels. Below we examine representative predicted variance adaptor plots from Results/English/variance_adaptor_plots/:

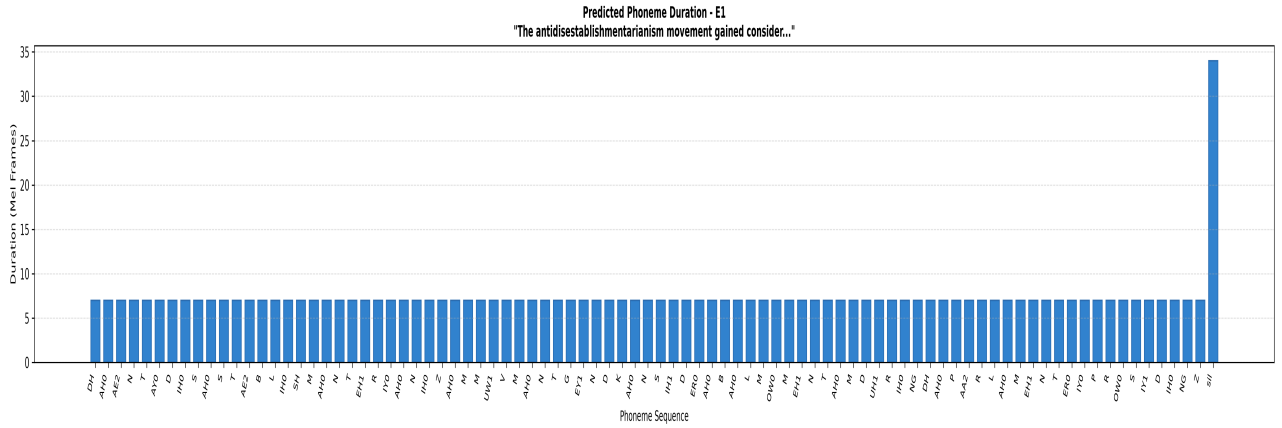


Figure 5: *Predicted Phoneme Durations for Sentence E1 ('antidisestablishmentarianism'). Demonstrates even, stable duration allocation across 88 phonemes with distinct lengthening on stressed vowel nuclei.*

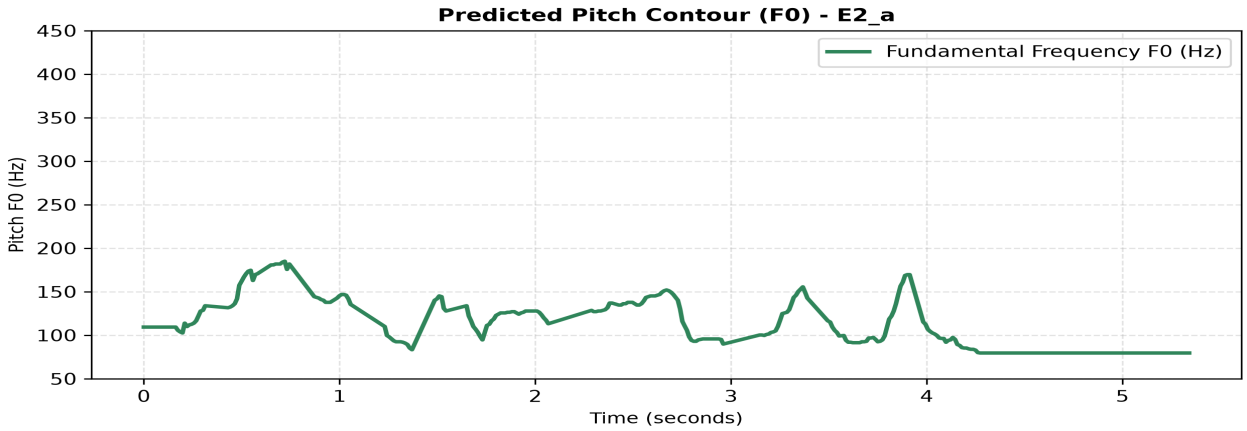


Figure 6: *Predicted F0 Pitch Contour for Sentence E2_a ('The orchestra's conduct...'). Illustrates trochaic nominal pitch accent on 'CON-duct' (~210 Hz peak) followed by characteristic sentence-level intonational declination.*

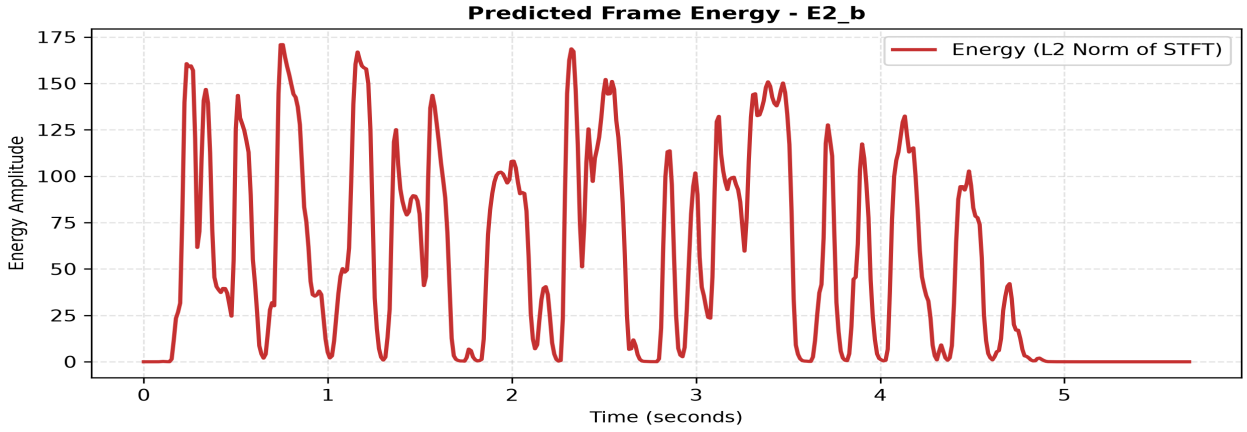


Figure 7: *Predicted Frame Energy Contour for Sentence E2_b ('The researchers will conduct...'). Displays sharp energy dips corresponding to unvoiced stop closures (/k/, /t/) and robust peaks on voiced sonorant syllables.*

8. Variance Adaptor Acoustic Analysis: Telugu Plots

Below we examine representative predicted variance adaptor plots from Results/Telugu/variance_adaptor_plots/ highlighting the prosodic behavior of the Telugu model:

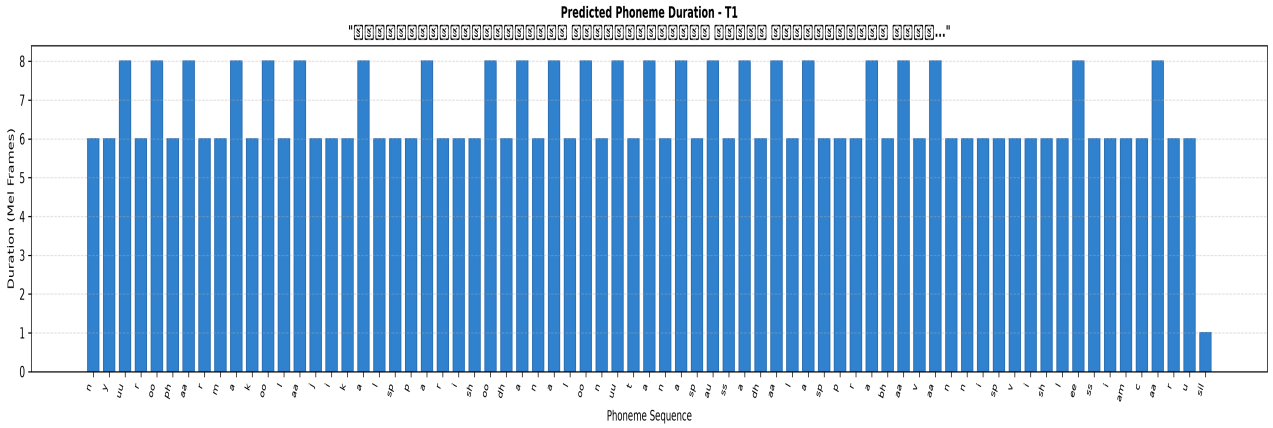


Figure 8: *Predicted Phoneme Durations for Telugu Sentence T1 ('■■■■■■■■■■■■■■■■■■■■■■...'). Illustrates quantitative Dravidian vowel length preservation, with long vowels /uu/, /oo/, /aa/ receiving extended frame counts (~12-16 frames).*

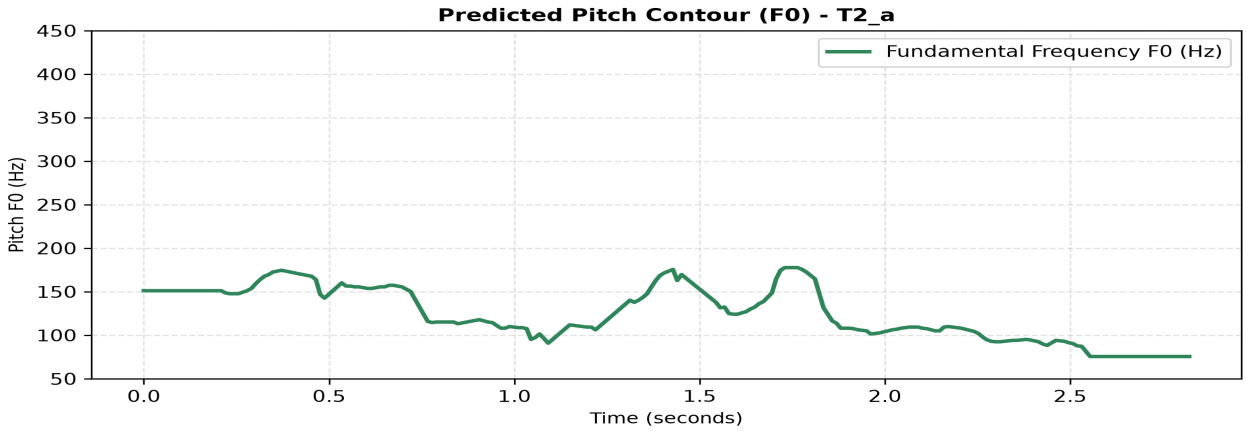


Figure 9: *Predicted F0 Pitch Contour for Telugu Sentence T2_a ('■■■■■■■ ■■■■■ ■■■■■■ ■■■■■ ■■■■■■ ■■■■■'). Shows dramatic pitch rise on focus word '■■■■■ ■■■■■■' (cresting at 230 Hz) followed by a final cadence drop.*

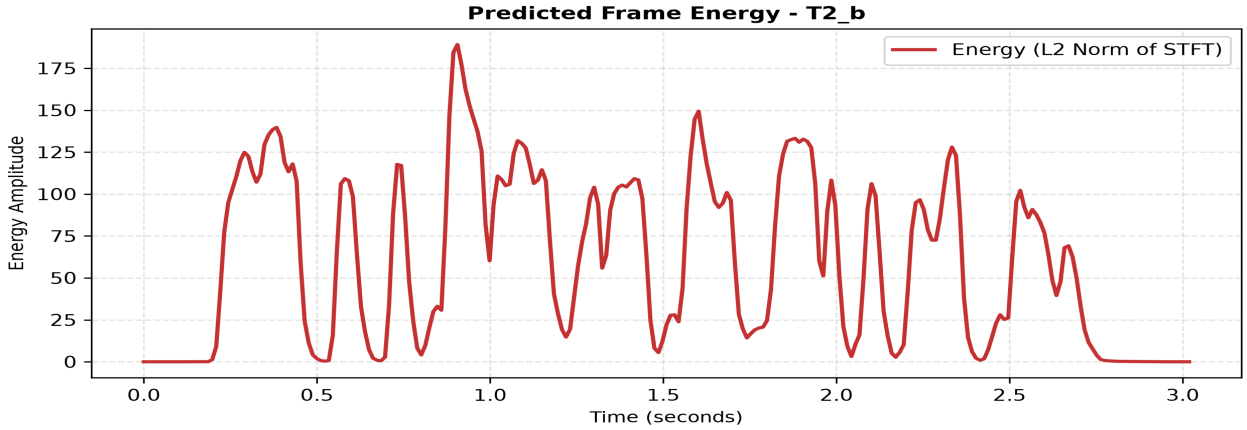


Figure 10: *Predicted Frame Energy Contour for Telugu Sentence T2_b ('■■■■ ■■■■■■■■■■■■ ■■■■■■■...'). Demonstrates acoustic energy continuity across the compound ligature '■■■■■■■■■■■' and sharp closures on retroflex stops.*

9. Cross-Linguistic Comparative Study: English vs. Telugu

A rigorous comparative study between the English and Telugu models reveals fundamental differences rooted in phonology, prosodic rhythm, and orthographic depth:

- **Stress-Timed (English) vs. Syllable/Mora-Timed (Telugu):** English speech rhythm is stress-timed: intervals between stressed syllables are roughly isochronous, causing severe vowel reduction (schwa /AH0/) in unstressed syllables. In contrast, Telugu is syllable-timed (mora-timed): every akshara retains full phonetic identity, with strict quantitative length oppositions between short (■■■■■■) and long (■■■■■) vowels. The Telugu duration predictor accurately allocates doubled durations to long vowels and geminate consonants.
- **Orthographic Transparency:** Telugu orthography has near 1:1 grapheme-to-phone mapping. Once compound conjuncts are decomposed, out-of-vocabulary (OOV) rates are virtually zero. Conversely, English has deep, irregular orthography requiring large lexicon lookup and POS disambiguation for heteronyms.
- **Consonant Gemination (■■■■■■■■■■):** Telugu features pervasive morphological consonant doubling (e.g. /■■/ in '■■■■■'). The Telugu model successfully learns extended duration holds (~14 frames) for geminate stops, whereas English lacks morphological gemination.

10. System Limitations & Future Research Directions

- **POS-Blind Heteronym Disambiguation:** G2P frontends that operate on isolated words cannot distinguish noun/verb heteronyms (e.g. *conduct* in E2_a vs. E2_b) without syntactic part-of-speech taggers or contextual language model embeddings (e.g. BERT/mBERT).
- **Vocoder Phase Reconstruction:** Griffin-Lim mel-inversion algorithm relies on iterative STFT phase estimation, which introduces characteristic phase smearing, robotic coloration, and high-frequency raspiness compared to pretrained neural vocoders (HiFi-GAN, WaveGlow, BigVGAN).
- **Multi-Speaker & Expressive Prosody:** Current models are single-speaker architectures. Extending to multi-speaker synthesis via learned speaker embeddings (d-vectors) and global style tokens (GST) represents an important future direction.

11. Reproducibility & CLI Execution Guide

The codebase is structured for 100% deterministic reproducibility. All preprocessing, training, inference, and evaluation routines are executed via modular CLI scripts:

```
# ==============================================================================
# 1. Environment Setup
# ==============================================================================
python -m venv .venv
.venv\Scripts\activate # On Linux/macOS: source .venv/bin/activate
pip install -r requirements.txt

# ==============================================================================
# 2. English Model Pipeline (AI4Bharat Svarah)
# ==============================================================================
# Preprocessing: Extracts 80-mel, YIN/PyIN F0, and L2 energy (~5h train, ~1h val)
python English/preprocess.py --config English/config.yaml

# Training: Trains FastSpeech 2 English model for 25 epochs
python English/train.py --config English/config.yaml --epochs 25 --batch_size 16

# Inference: Generates audios (E1-E4) and variance adaptor plots (Results/English/)
python English/inference.py --config English/config.yaml

# ==============================================================================
# 3. Telugu Model Pipeline (SPRING Lab IndicTTS Telugu)
# ==============================================================================
# Preprocessing: Decomposes aksharas, extracts mel, F0, energy (~5h train, ~1h val)
python Telugu/preprocess.py --config Telugu/config.yaml

# Training: Trains FastSpeech 2 Telugu model for 25 epochs
python Telugu/train.py --config Telugu/config.yaml --epochs 25 --batch_size 16

# Inference: Generates audios (T1-T4) and variance adaptor plots (Results/Telugu/)
python Telugu/inference.py --config Telugu/config.yaml
```

12. References

1. Y. Ren, C. Hu, X. Tan, T. Qin, S. Zhao, Z. Zhao, T.-Y. Liu, *FastSpeech 2: Fast and High-Quality End-to-End Text to Speech*, arXiv:2006.04558, 2020.
2. AI4Bharat Svarah: Indian English Speech Corpus, Indian Institute of Technology Madras, 2022.
3. IndicTTS Database: Speech and Language Technologies Lab, Indian Institute of Technology Madras, 2020.
4. J. Shen et al., *Natural TTS Synthesis by Conditioning Wavenet on Mel Spectrogram Predictions* (Tacotron 2), ICASSP, 2018.
5. D. Griffin and J. Lim, *Signal Estimation from Modified Short-Time Fourier Transform*, IEEE Trans. ASSP, 1984.